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The differentiated effect of NO and NO_2 in promoting methane oxidation

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Abstract

It has long been recognised that nitric oxide (NO) and nitrogen dioxide (NO_2) can promote the oxidation of methane and other hydrocarbons under various fuel-lean and fuel-rich conditions and a range of temperatures and pressures conditions. However, despite the ease with which NO and NO_2 interconvert under the oxidation conditions, the reactions responsible for initiating the oxidation process for methane are different. This is demonstrated experimentally by the different temperature characteristics of the sensitised reaction and supported by kinetic modelling. In this study, the effect of NO/NO_2-sensitised oxidation of methane was experimentally studied with two fuel-lean mixtures, viz. 2.5% methane-in-air and 0.05% methane-in-air mixtures, at atmospheric pressure, over temperature ranges of 823–948 K, and 873–1023 K, respectively. Kinetic modelling of the experimental results showed satisfactory agreement and an exhaustive sensitivity analysis was conducted. It was shown through brute-force sensitivity analysis that the difference in key reactions in both NO and NO_2 system leads to the observed trend: the effect of NO starts at lower temperatures while NO_2 appears to be more potent at higher temperatures. Another salient point emerging from the sensitivity analysis is that the initiation reaction in NO-promoted system was identified to be $CH_3O_2 + NO = CH_3O + NO_2$; for NO_2-sensitised case, $CH_4 + NO_2 = CH_3 + HONO$ is the most important initiator at lower temperatures while $NO_2 + O = NO + O_2$ and $NO_2 + HO_2 = HONO + O_2$ appear to be of prime importance at higher temperatures.
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1. Introduction

It has been known since the 1920's [1,2] that the addition of small amounts of NO_x gases (NO and NO_2) can promote the oxidation of hydrocarbons. There are numerous more recent publications dealing with experimental and modelling studies related to the interactions of hydrocarbons and nitrogen oxides [3–28]. Most of these studies consider fuel-lean conditions, though the propensity of NO_x to promote the formation of methanol as a product of partial oxidation of methane has also led to a smaller extent of investigations of the interaction under fuel-rich conditions [10,13,14,16,17,23]. Similarly, while the original studies centred on

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operating pressures close to atmospheric, more recent investigations include effects at elevated pressures [20,22,24–26]. A significant aspect of the above promotion where NO is the agent, is that it is mutual: while the oxidation of hydrocarbons is accelerated, at least a fraction of the NO present is oxidised to NO_2 [5–7,18,19,21].

The mechanism of the NO_x promotion has been studied using both experimental and modelling techniques for a range of fuel gases [9,11,12,15,18–22,26–28]. However, because NO and NO_2 are readily interconverted and may "recycle" [15,26], their "native" roles in the promoting effect has not been unambiguously elucidated. It is apparent that, while both species are capable of promoting the oxidation of hydrocarbons (and other combustible gases such as hydrogen and carbon monoxide [2,12]) under certain conditions, their efficacy may differ.

For instance, it was found [5] that NO strongly promoted the oxidation of *n*-butane (approximately 45 ppm v/v) in air at a temperature of about 650 K, while NO_2 was almost completely ineffective. Likewise, Lewis and von Elbe [2] concluded that NO can readily promote the oxidation of hydrogen. They suggested that the observed promoting effect of NO_2 (and NOCl) is the result of NO formation from these species. Mueller et al. [12] also reported that the promoting effect of NO_2 on hydrogen and carbon monoxide oxidation is due to NO formation. In contrast, Slack and Grillo [4] found in shock tube experiments that NO_2 was a strong promoter of methane–oxygen ignition, while NO was not. It has also been observed that NO can be effective in promoting hydrocarbon oxidation when present in the parts per million concentrations [8] and at even lower levels [5].

Bendtsen et al. [15] extensively studied the effects of NO and NO_2 in promoting methane oxidation under fuel-lean conditions in a flow reactor and their results showed that both species were promoters in the temperature range of 800–1000 K. There was a definite tendency for the reaction to begin at lower temperatures with NO as the promoter. However, as the temperature was raised NO_2 became increasingly more effective and its promoting effect eventually overtook that of NO. This is well illustrated in Fig. 1, the data for which was extracted from Bendtsen et al. [15]. The relevant experiment was conducted under atmospheric pressure at residence time of approximately 0.2 s and fuel-lean conditions (averaging around 2250 ppm methane and 3.6% oxygen with nitrogen the balance). NO or NO_2 added in the amounts close to 200 ppm clearly differ in promoting efficiency at different temperatures. In relation to this figure, and subsequently, we define the "crossover temperature" as the point at which the promoting effect of NO_2 equals that of NO. It may be noted that Bendtsen et al. [15] proposed a mechanism for the NO_x-promoted oxidation of methane,

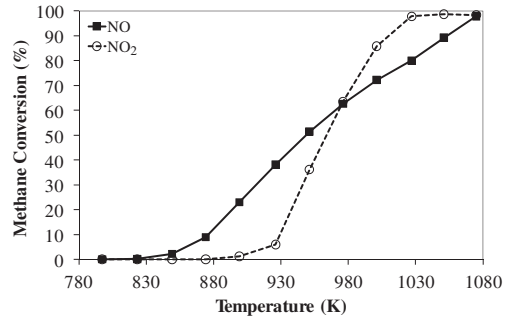

Fig. 1. 1 NO/NO_2 sensitisation crossover derived from experimental data of Bendtsen et al. [15].

showing alternative channels for NO and NO_2 activation, which also result in cycling of these species. Specifically, they identified the critical reactions as $CH_3O_2 + NO = CH_3O + NO_2$ and $CH_3 + NO_2 = CH_3O + NO$.

Takemoto et al. [17], in studying the promoting effects of NO_x under fuel-rich conditions, cited theoretical studies that show the initiation reaction of methane with NO_2 ($CH_4 + NO_2 = CH_3 + HONO$) has a considerably lower activation energy than the equivalent reaction with NO ($CH_4 + NO = CH_3 + HNO$), respectively, 191.76 vs 274.65 kJ/mol. They nevertheless noted that NO was at least as effective in promoting the oxidation of methane under the fuel-rich conditions of their experiments and in the temperature region around 750 K as was NO_2. Zalc et al. [23] found NO_2 to act more rapidly as a promoter under very fuel-rich conditions than did NO, and Rasmussen et al. [26] also noted the lower energy barrier of the above NO_2 reaction.

Dagaut et al. [19,20,22,24,25,28] have recently published comprehensive reaction sets dealing with the NO_x promotion of hydrocarbon oxidation, including at elevated pressures. Their model was found to well represent our data and is used herein.

In this work we examine the roles of NO and NO_2 as promoters in atmospheric, fuel-lean conditions through experiments and kinetic modelling. Sensitivity analysis was attempted to gain a better insight into the kinetics of NO and NO_2 promotion, particularly the initiation of the reaction.

2. Experimental details

The experiments were conducted in a laminar-flow reactor, which was housed in an electrically-heated and fan-stirred oven that provided a constant and uniform temperature profile [9]. The laminar-flow reactor consisted of an uncoated Vycor tube with inner diameter of 16.35 mm and total length of 2.44 m, in which three off-take

points were evenly spaced, with off-takes 1, 2 and 3 located at 0.82, 1.64 and 2.44 m, respectively, enabling samples to be extracted at different residence times. This reactor effectively simulated plug flow and further details of the apparatus have been described elsewhere [9].

In this contribution, two sets of experimental data are presented, both conducted at atmospheric pressure. The first experiment studied the separate effects of 0–100 ppm of NO and NO_2 in a 2.5% (25,000 ppm) methane-in-air mixture over the temperature range of 823–948 K at residence time of approximately 2 s (total feed flow rate of 5000 mL min^{-1} and samples withdrawn from off-take 3). In the second experiment, the individual effects of 0–100 ppm of NO and NO_2 were investigated in a 0.05% (500 ppm) methane-in-air mixture over the temperature range of 873–1023 K at a relatively shorter residence time of 0.35 s (total feed flow rate of 10,200 mL min^{-1} and samples withdrawn from off-take 1).

In both experiments, the reaction mixture, which was made up from separately metered streams of CO_2-free air and methane, was introduced into the reactor via a Vycor preheating section located within the kiln. NO/NO_2 (in N_2) was added immediately before the reactor. In blank runs without NO addition, minimal reaction occurred over the full length of the active reactor, demonstrating that any reaction in the preheating zone was negligible. The flow rates of the feed, which summed to either 5000 or 10,200 mL min^{-1} (standard condition), were determined using a calibrated flow meter.

All major stable species exiting the reactor were quantified by using online monitoring devices. The concentrations of unconverted reactant (methane) and the products formed (carbon oxides) were determined by Gas Chromatography (GC). The GC used in this work employed a capillary GS-Q column and was equipped with a methaniser with efficiency approaching 100%. The accuracy of the GC analysis results was estimated to be within ±5%. In addition to GC analysis, the concentration of carbon oxides was also measured by using non-dispersive infrared (NDIR) analyser. On the other hand, the nitrogen oxide concentrations were determined using a conventional chemiluminescence analyser equipped with a stainless-steel converter. For all the analytical equipment used in this work, the calibration standards for CH_4, CO, CO_2 and NO_x were traceable to primary gravimetric standards. The carbon balance was typically within 5% (i.e. between 95% and 105% of the carbon entering the reactor was found exiting).

3. Kinetic modelling

In this work, kinetic modelling was performed with Chemkin 4.1, using a plug flow reactor model, assuming isothermal conditions at atmospheric pressure. The kinetic mechanism of Le Cong et al. [24,25], which has been validated over a wide range of operating conditions (mostly in the temperature range of 900–1500 K for flow reactors) and was also found to perform better than other models in the previous work of the authors [27], was used. Brute-force sensitivity analysis was employed to identify the most important rate-controlling reactions. It may be noted that the brute-force method provides a global sensitivity from the initiation time to the time when maximum concentration of OH radical is reached. In addition, Chemkin Pro was used to calculate the localised sensitivity coefficients (targeting CH_4 concentration) at given points along the reactor (effectively residence times), which allowed the investigation of important reaction(s) that initiates the system in both NO and NO_2 cases.

4. Results and discussion

Figure 2 illustrates the experimental results (for both 2.5% methane-in-air mixture and 0.05% methane-in-air mixture) of NO_x promotion on methane conversion, associated with their respective modelling results simulated using the kinetic model from Le Cong et al. [24,25]. It may be seen that, the model from Le Cong et al. [24,25] performs reasonably well in reproducing the experimental trends: NO initiates the reaction at a much lower temperature (than NO_2) while the sensitising effect of NO_2, once commenced, increases more rapidly than that of NO with rising temperature. At higher temperatures, the sensitising effect of NO_2 overtakes that of NO, as may be seen from the experimental data presented. With the satisfactory agreement achieved between the experimental and modelling results, further sensitivity analysis was carried out, attempting to reveal the important reactions that cause the difference between NO and NO_2 effects. In this work, sensitivity analysis was only conducted with the experimental conditions where 0.05% methane-in-air mixture was used for the following two reasons: (a) the difference between NO and NO_2 effects was found to be more significant at lower methane concentration and shorter residence time and (b) greater heat generation will occur in the oxidation of the 2.5% methane-in-air mixture than would be the case at 0.05%. This may affect the accuracy of the calculation based on the assumption of isothermal conditions, although the reactor geometry is expected to mitigate this effect to a considerable extent. Nevertheless, the results obtained with the 0.05% methane-in-air mixture are more suitable for modelling under isothermal conditions and quantitative analysis purposes, as the heat generated in that case is not a significant consideration.

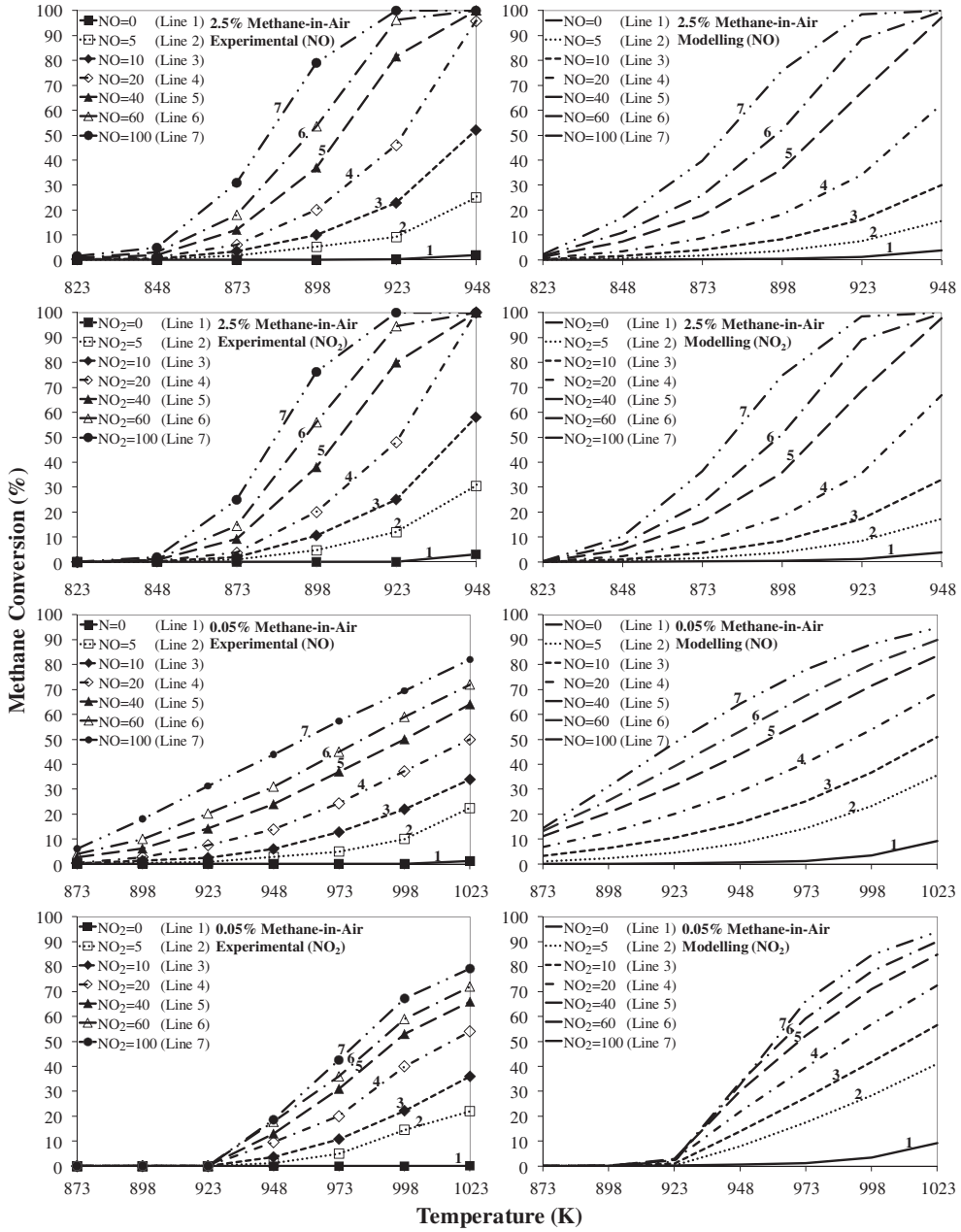

Fig. 2. Experimental and modelling results of NO/NO_2 effects on methane conversion. (Experiment with 2.5% methane-in-air mixture: 0–100 ppm NO/NO_2 added, over temperature range of 823–948 K, at atmospheric pressure and residence time of 2 s. Experiment with 0.05% methane-in-air mixture: 0–100 ppm NO/NO_2 added, over temperature range of 873–1023 K, at atmospheric pressure and residence time of 0.35 s.)

Figure 3 shows the brute-force sensitivity analysis results for the 0.05% methane-in-air mixture sensitised by 40 ppm of NO and NO_2. The reactions appeared in Fig. 3 are listed in Table 1. In Fig. 3, the sensitivity coefficients of the top 10 most important reactions were plotted as a function of temperature (923–1023 K). In this work, the brute-force sensitivity coefficient profile only applies to temperatures above 923 K because there was virtually no reaction observed below this temperature with the case that NO_2 was added as promoter (Fig. 2). It may be seen from Fig. 3 that the sensitivity coefficient profiles for both NO and NO_2 cases are quite different. As observed from the left column of Fig. 3, in the case that NO was added, the sensitivity coefficient

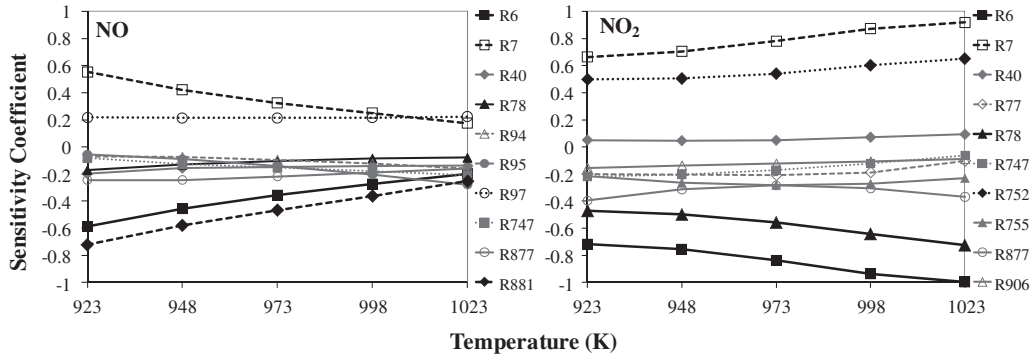

Fig. 3. Brute-force sensitivity coefficient profile as a function of temperature in 0.05% methane-in-air mixture sensitised by 40 ppm NO (left) or NO_2 (right).

Table 1
Important reactions in the oxidation of 0.05% methane-in-air mixture sensitised by 40 ppm NO_x.

Reaction details	
R6	$H + O_2 = OH + O$
R7	$H + O_2 + M = HO_2 + M$
R40	$CH_2O + OH = HCO + H_2O$
R77	$CH_4 + OH = CH_3 + H_2O$
R78	$CH_4 + O = CH_3 + OH$
R94	$CH_3 + O_2 = CH_2O + OH$
R95	$CH_3 + O_2 = CH_3O + O$
R97	$CH_3 + CH_3(+M) = C_2H_6(+M)$
R747	$NO + HO_2 = NO_2 + OH$
R752	$NO_2 + O = NO + O_2$
R755	$NO_2 + HO_2 = HONO + O_2$
R877	$CH_3 + NO_2 = CH_3O + NO$
R881	$CH_3O_2 + NO = CH_3O + NO_2$
R906	$CH_3 + HONO = CH_4 + NO_2$

for the top four most important rate-controlling reactions (i.e. R6, R7, R97 and R881) mostly reduces with increasing temperature, except R97 that remains almost independent of temperature. On the other hand, in the case that NO_2 was used as promoter, the sensitivity coefficient for the top four key reactions (i.e. R6, R7, R78 and R752) increases with increasing temperature. In addition to the opposing trends for sensitivity coefficient profile, the 10 most important reactions vary considerably in both NO and NO_2 cases. For instance, R881 is of greatest importance throughout the temperature range with NO but it does not feature with NO_2 promotion. Also, R752 which appears to be very important in the NO_2 case, is not in the list of the 10 most significant reactions for the NO case. It is apparent that the variation in sensitive reactions in both NO and NO_2 cases lead to the different characteristics in the promoting effects of NO and NO_2.

In addition to the brute-force sensitivity analysis, in which sensitivity coefficients were taken at the maximum [OH] (in other words, maximum reaction rate), localised sensitivity analysis coefficients along the reactor were also calculated using Chemkin Pro, for the 0.05% CH_4 in air for sensitisation with 100 ppm of NO and NO_2. Figures 4 and 5 shows the normalised sensitivity coefficient of CH_4 for the five most important reactions at initiation (methane conversion between 0.10% and 0.14%) sensitised by NO and NO_2, respectively, at temperatures of 923 K (lower temperature limit at which sufficient methane conversion is achieved to assure reliable sensitivity analysis results) and 1023 K (maximum temperature studied in this work).

Figures 4 and 5 identify the very different key reactions at the initiation for NO vs NO_2 sensitisation. As seen from Fig. 4, for NO-sensitised system, reaction $CH_3O_2 + NO = CH_3O + NO_2$ (R881) accelerates methane oxidation by converting relatively unreactive methylperoxy to methoxy radicals, at both temperatures of 923 and 1023 K and this is also observed for all other temperatures investigated in this study. Further sensitivity analysis (not presented herein) showed the significance of R881 to weaken as the reaction proceeds along the reactor; this may be attributed to the generation of a pool of more reactive radicals. Following the decline of the importance of R881, reactions $CH_3 + NO_2 = CH_3O + NO$ (R877), $NO_2 + O = NO + O_2$ (R752) and $NO + HO_2 = NO_2 + OH$ (R747) then emerged as the important NO_x-related reactions.

On the other hand, for the NO_2-promoted system (Fig. 5), at a lower temperature of 923 K, the reverse of reaction $CH_3 + HONO = CH_4 + NO_2$ (R906) initiates the conversion of methane. However, at higher temperatures (above 998 K), the initiation of a NO_2-sensitised system differs from that at low temperatures. As noticed from Fig. 5, at temperature of 1023 K, reaction $NO_2 + O = NO + O_2$ (R752) and $NO_2 + HO_2 = HONO + O_2$ (R755) are the two important NO_x-related reactions. R755 contributes strongly to the acceleration of the overall oxidation. This is attributed to the rapid decomposition of the product HONO into active radicals. Besides, it is also worthwhile to note that

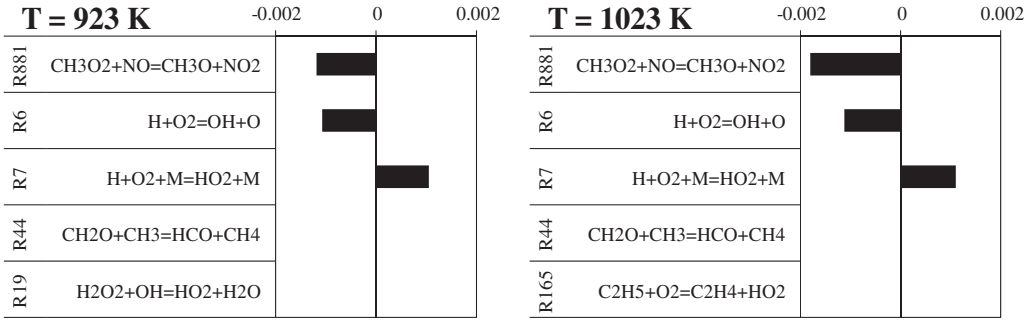

Fig. 4. Normalised CH_4 sensitivity coefficient of the five most important reactions at initiation sensitised by 100 ppm NO at temperatures of 923 and 1023 K (Left: $T = 923$ K, sensitivity coefficients extracted when methane conversion = 0.10%. Right: $T = 1023$ K, sensitivity coefficients extracted when methane conversion = 0.14%).

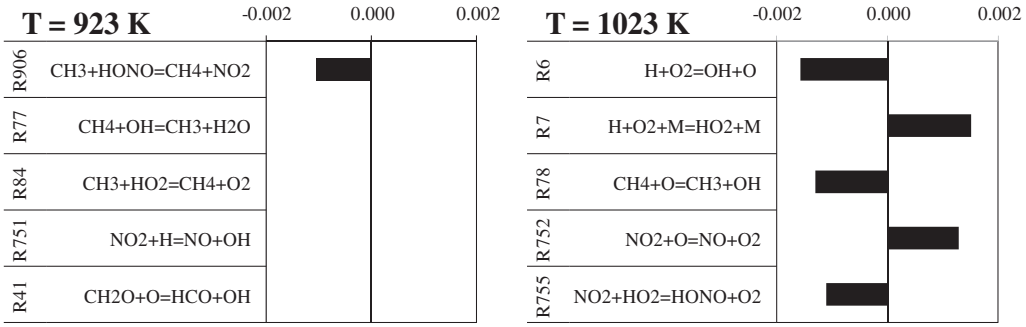

Fig. 5. Normalised CH_4 sensitivity coefficient of the five most important reactions at initiation sensitised by 100 ppm NO_2 at temperatures of 923 and 1023 K (Left: $T = 923$ K, sensitivity coefficient extracted when methane conversion = 0.12%. Right: $T = 1023$ K, sensitivity coefficient extracted when methane conversion = 0.10%).

in the CH_4/air/NO_2 system, R752 is an important reaction even at low methane conversions (0.15% compared to 2-3% for the CH_4/air/NO system).

A further aspect of significance in differentiating the roles of NO and NO_2 relates to their effectiveness at different temperatures. Inspection of Fig. 2 shows that the "crossover temperature", the temperature at which NO_2 produces an equal sensitising effect to NO, rises with increasing NO_x concentration. This is also reflected in both the experimental and modelling data as summarised in Fig. 6, where methane concentrations of 100–25,000 ppm are modelled and experimental points are given for the 500 and 25,000 ppm cases.

The "crossover temperature" varies with both concentrations of methane and of NO_x. The very existence of the "crossover temperature" can be understood by recalling the variation of the brute-force sensitivities as a function of temperature (at fixed concentrations of CH_4 and NO_x, as shown in Fig. 3). Key reactions in the mixtures with NO have decreasing sensitivities with increasing temperature, while those in the mixtures with NO_2 increase. The total fluxes via these most sensitive reactions thus inevitably cross at some point; the "crossover temperature". At higher concentrations of methane, the variation of the

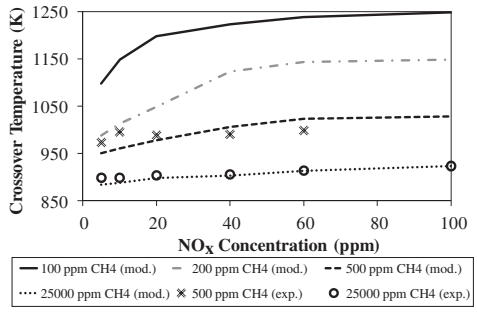

Fig. 6. Crossover temperature as a function of NO_x concentration.

"crossover temperature" with NO_x level weakens. This is due to the fact that at greater concentrations of methane, the levels of the chain-leading radicals are higher and therefore less sensitive to the amount of the NO_x added. At smaller concentrations of CH_4 the levels of these radicals are much lower, which leads to a reduced role of the radical–radical reactions. Note that R97, which is important in the mixtures with NO added, has a lesser temperature dependence and is chain-terminating (Fig. 3). Partial suppression of this reaction makes the mixtures with NO added more

reactive at lower temperatures compared to NO_2-sensitised mixtures. It follows that higher temperatures are required for NO_2-sensitised mixtures to reach the same level of reactivity as their NO-sensitised counterparts (hence the existence of the "crossover temperature").

It should also be noted that the case of "small concentrations of CH_4 (example,100 ppm) with small concentrations of NO_x", as represented by the top line (solid line) in Fig. 6, is essentially a limiting condition between oxidation of methane promoted by traces of NO_x and oxidation of NO promoted by traces of methane. The chemistry governing this case has been discussed elsewhere [29] and is beyond the scope of this study.

5. Conclusions

The separate effects of NO and NO_2 in methane oxidation has been experimentally studied in two fuel-lean mixtures (viz. 2.5% methane-in-air mixture and 0.05% methane-in-air mixture) in an atmospheric flow reactor. It has been confirmed that the promotion by NO commences at lower temperatures than with NO_2 and the latter becomes more potent at higher temperatures. Kinetic modelling using the mechanism from Le Cong et al. demonstrated good agreement with the experimental trend. Brute-force sensitivity analysis revealed that the variation of important rate-controlling reactions in both NO and NO_2-promoted systems are responsible for the different sensitising behaviours shown by NO and NO_2. Analysis of the localised sensitivity coefficient profile indicated that initiation reaction in the NO-promoted environment is: $CH_3O_2 + NO = CH_3O + NO_2$. This is in total contrast to the situation with NO_2 promotion where the reaction $CH_4 + NO_2 = CH_3 + HONO$ dominates over low temperatures while at higher temperatures (above 998 K), $NO_2 + O = NO + O_2$ and $NO_2 + HO_2 = HONO + O_2$ appear to be the most important NO_x-related reactions.

Acknowledgements

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References

[1] C.H. Bibb, H.J. Lucas, *Ind. Eng. Chem.* 21 (7) (1929) 633–638.
[2] B. Lewis, G. von Elbe, *Combustion, Flames and Explosions of Gases*, Academic Press Inc., Orlando, Florida, 1987.
[3] R.F. Sawyer, I. Glassman, *Proc. Combust. Inst.* 12 (1969) 469.
[4] M.W. Slack, A.R. Grillo, *Combust. Flame* 40 (1981) 155–172.
[5] J.H. Bromly, *The Formation of Nitrogen Oxides in Gas Combustion*, Ph.D. dissertation, Department of Engineering, Murdoch University, Perth, Western Australia, 1991.
[6] F.J. Barnes, J.H. Bromly, R. Mandyczewsky, T.J. Edwards, B.S. Haynes, in: *International Gas Research Conference, The Gas Research Institute*, Orlando, Florida, 1992.
[7] J.H. Bromly, F.J. Barnes, R. Mandyczewsky, T.J. Edwards, B.S. Haynes, *Proc. Combust. Inst.* 24 (1993) 899–907.
[8] S.K. Prabhu, H. Li, D.L. Miller, N.P. Cernansky, *SAE Tech. Paper Ser.* 932757 (1993) 1–7.
[9] J.H. Bromly, F.J. Barnes, S. Muris, X. You, B.S. Haynes, *Combust. Sci. Technol.* 115 (1996) 259–296.
[10] T. Amano, F.L. Dryer, *Proc. Combust. Inst.* 27 (1998) 397–404.
[11] M. Jazbec, B.S. Haynes, J.H. Bromly, F.J. Barnes, in: *Proceedings of the Second Asia-Pacific Conference on Combustion*, The Combustion Institute, Tainan, Taiwan, 1999, pp. 158–161.
[12] M.A. Mueller, R.A. Yetter, F.L. Dryer, *Int. J. Chem. Kinet.* 31 (10) (1999) 705–724.
[13] K. Otsuka, R. Takahashi, I. Yamanaka, *J. Catal.* 185 (1) (1999) 182–191.
[14] Y. Yamaguchi, Y. Teng, S. Shimomura, K. Tabata, E. Suzuki, *J. Phys. Chem. A* 103 (41) (1999) 8272–8278.
[15] A.B. Bendtsen, P. Glarborg, K. Dam-Johansen, *Combust. Sci. Technol.* 151 (2000) 31–72.
[16] K. Tabata, Y. Teng, Y. Yamaguchi, H. Sakurai, E. Suzuki, *J. Phys. Chem. A* 104 (12) (2000) 2648–2654.
[17] T. Takemoto, K. Tabata, Y. Teng, A. Nakayama, E. Suzuki, *Appl. Catal. A* 205 (1-2) (2001) 51–59.
[18] T. Faravelli, A. Frassoldati, E. Ranzi, *Combust. Flame* 132 (2003) 188–207.
[19] P. Dagaut, O. Mathieu, A. Nicolle, G. Dayma, *Combust. Sci. Technol.* 177 (2005) 1767–1791.
[20] P. Dagaut, A. Nicolle, *Combust. Flame* 140 (2005) 161–171.
[21] P.A. Glaude, N. Marinov, Y. Koshiishi, N. Matsunaga, M. Hori, *Energy Fuels* 19 (2005) 1839–1849.
[22] P. Dagaut, G. Dayma, *J. Phys. Chem. A* 110 (2006) 6608–6616.
[23] J.M. Zalc, W.H. Green, E. Iglesia, *Ind. Eng. Chem. Res.* 45 (8) (2006) 2677–2688.
[24] T. Le Cong, P. Dagaut, *Combust. Sci. Technol.* 180 (2008) 2046–2091.
[25] T. Le Cong, P. Dagaut, G. Dayma, *J. Eng. Gas Turbines Power* 130 (4) (2008) 41502–41511.
[26] C.L. Rasmussen, A.E. Rasmussen, P. Glarborg, *Combust. Flame* 154 (2008) 529–545.
[27] Y.L. Chan, J.H. Bromly, A.A. Konnov, D.K. Zhang, in: *Australian Combustion Symposium*, The Combustion Institute, Brisbane, Australia, 2009.
[28] G. Dayma, P. Dagaut, in: *4th European Combustion Meeting*, The Combustion Institute, Vienna, Austria, 2009.
[29] M. Hori, N. Matsunaga, N. Marinov, W. Pitz, C. Westbrook, *Proc. Combust. Inst.* 27 (1998) 389–396.